§3.5 Fermi气体

Fermi子：电子、质子、中子、中微子、Majorana费米子

简单历史

- **1900年，Plank量子假设. Max Plank, 1918 Nobel Prize, in Physics**

  "for the services he rendered to the advancement of physics by his discovery of energy quanta"

- **1924年，Bose子、Bose-Einstein统计.**

- **1925年，Pauli不相容. Wolfgang Pauli, 1945 Nobel Prize in Physics，**

  for his "decisive contribution through his discovery of a new law of Nature, the exclusion principle or Pauli principle."

- **1926年，Fermi子、Ferm-Dirac统计.**

  Enrico Fermi, 1938 Nobel Prize in physics, "for his demonstrations of the existence of new radioactive elements produced by neutron irradiation, and for his related discovery of nuclear reactions brought about by slow neutrons".

  Paul Dirac, 1933 Nobel Prize in physics, for the discovery of new productive forms of atomic theory".

- **1926年, Sir Ralph Howard Fowler, Collapse of a star to White Dwarf**

(Paul Dirac's advisor, introduces Dirac to quantum theory in 1923. S. Chandrasekhar, Nobel Prize in 1983, Nevill Francis Mott, Nobel Prize in 1977).

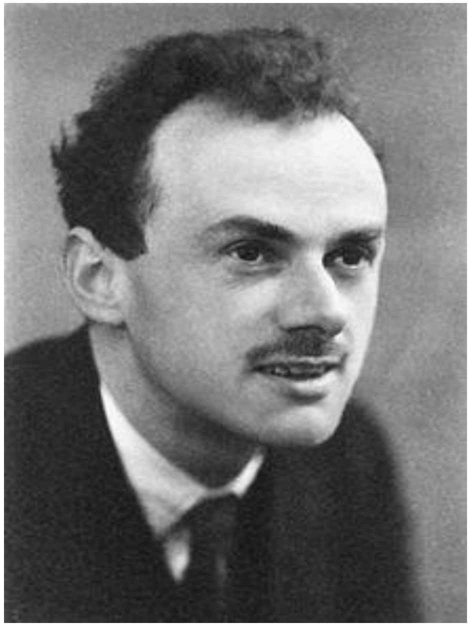
- **1927年, Arnold Sommerfeld, Drude-Sommerfeld Model.**

Four of his students, Werner Heisenberg (1932), Wolfgang Pauli (1945), Peter Debye (1936), and Hans Bethe (1967) went on to win Nobel Prizes. Linus Pauling (1954, chemistry and 1962, peace), Herbert Kroemer (2000) also studied under him at some point.

- **1928年, Fowler and Lothar Wolfgang Nordheim.**

Field Electron Emission from metals, or called FN tunneling.





Wolfgang Pauli
(Switzerland)
1900–1958

Enrico Fermi
(Italy)
1901–1954

Paul Dirac
(England)
1902–1984

服从Fermi分布的系统和Boltzmann系统及Bose系统的差别都很大. 特别是电子“气体”, 经常遇见的情形是 $n\lambda^3 \gg 1$ 即 ($e^{\alpha} \ll 1$), 所以必须按量子气体来处理.

一、绝对零度的情形($T = 0\,\mathrm{K}$)

- 完全简并

$$T = 0\,\mathrm{K},\ \lambda = \frac{h}{\sqrt{2\pi mkT}} \to \infty, \quad n\lambda^3 = n\left(\frac{h^2}{2\pi mkT}\right)^{3/2} \to \infty$$

一种理想情形, 称为完全简并

- 分布规律及物理意义

Fermi分布

$$n_i = \frac{g_i}{e^{\alpha+\beta\varepsilon_i} + 1}$$

引入单个量子态的平均粒子数(即是量子态被占据的几率)

$$f_i = \frac{n_i}{g_i} = \frac{1}{e^{\alpha+\beta\varepsilon_i} + 1}$$

$T \to 0\,\mathrm{K}$

$$\alpha + \beta\varepsilon_i = \frac{\varepsilon_i - \mu}{kT} \to \begin{cases} +\infty, & \varepsilon_i > \mu_0 \\ -\infty, & \varepsilon_i < \mu_0 \end{cases}$$

$$\mu_0 = \mu(T, V, N)\,|_{T=0}$$

μ_0：$T = 0\,\mathrm{K}$时系统的化学势.

$$e^x \to \begin{cases} +\infty, & x \to +\infty \\ 0, & x \to -\infty \end{cases}$$

$$\boxed{\lim_{T\to 0} f_i = \lim_{T\to 0} \frac{1}{e^{\frac{\varepsilon_i - \mu}{kT}} + 1} = \begin{cases} 1, & \varepsilon_i \le \mu_0 \\ 0, & \varepsilon_i > \mu_0 \end{cases}}$$

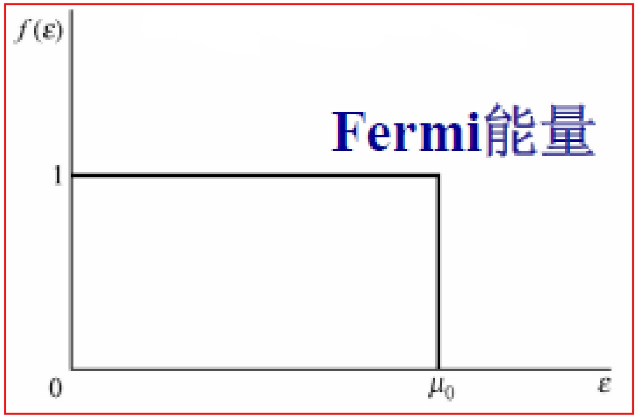


图3-5-1

这意味着, 低于μ_0的能级全被填满, 而高于μ_0的能级全部空着. 显然 μ_0 的值有极重要的意义.

● Fermi能量

μ_0称为系统的Fermi能量, 又记为 ε_F

对于非相对论粒子, 考虑平动的准连续形式, 在$\varepsilon - \varepsilon + d\varepsilon$之间的状态数

$$g(\varepsilon)d\varepsilon = \frac{2\pi VJ(2m)^{3/2}}{h^3}\sqrt{\varepsilon}d\varepsilon$$

$$\equiv CV\sqrt{\varepsilon}d\varepsilon$$

$$（C = \frac{2\pi J(2m)^{3/2}}{h^3}, J = 2S + 1）$$

系统的总粒子数

$$N = \sum_i n_i = \int_0^\infty f(\varepsilon)g(\varepsilon)d\varepsilon$$

$$= CV\int_0^{\mu_0}\sqrt{\varepsilon}d\varepsilon$$

$$= \frac{2}{3}CV\mu_0^{3/2}$$

$$\mu_0 = \left(\frac{3N}{2CV}\right)^{2/3} = \frac{h^2}{2m}\left(\frac{3}{4\pi}\frac{N}{JV}\right)^{2/3}$$

对于电子$S = \frac{1}{2}, J = 2$，电子气在绝对零度的化学势为

$$\mu_0 = \left(\frac{3N}{2CV}\right)^{2/3} = \frac{h^2}{2m}\left(\frac{3N}{8\pi V}\right)^{2/3}$$

$$\varepsilon_F = \mu_0$$

● 零点能量

$T=0$ 时Fermi气体的能量平均值

$$\bar{E}_0=\sum_i n_i\varepsilon_i=\int_0^\infty \varepsilon f(\varepsilon)g(\varepsilon)d\varepsilon$$

$$=\int_0^{\mu_0} CV\varepsilon^{3/2}d\varepsilon$$

$$=\frac{2}{5}CV\mu_0^{5/2}$$

$$\boxed{\bar{E}_0=\frac{3}{5}N\mu_0} \qquad (N=\frac{2}{3}CV\mu_0^{3/2})$$

费米气中单个粒子的平均能量

$$\bar{\varepsilon}_0=\frac{\bar{E}_0}{N}=\frac{3}{5}\mu_0$$

● 零点压强

$T=0\,\mathrm{K}$时,自由能$\bar{F}_0=\bar{E}_0\text{-T}\bar{S}_0=\bar{E}_0$,所以压强为

$$\bar{P}_0=-\left(\frac{\partial\bar{F}_0}{\partial V}\right)_T=-\left(\frac{\partial\bar{E}_0}{\partial V}\right)=\frac{2}{3}\frac{\bar{E}_0}{V}$$

$$=\frac{2}{5}\frac{N\mu_0}{V}$$

$$(\bar{E}_0=\frac{2}{5}\mathrm{CV}\mu_0^{5/2}\sim V^{-2/3},\ \mu_0=\frac{h^2}{2m}\left(\frac{3}{4\pi}\frac{N}{JV}\right)^{2/3})$$

数量级

电子: $m_e\sim 9.1\times10^{-31}\,\mathrm{kg}$

金属中电子的数密度: $n=\frac{N}{V}\sim 4\times10^{28}\,\mathrm{m}^{-3}$

$$\mu_0\sim\mathbf{6.9\times10^{-19}\,J},\ \bar{\varepsilon}_0\sim\mathbf{4\times10^{-19}\,J}$$

$$\bar{P}_0\sim\mathbf{10^{10}\,Pa}\sim\mathbf{10^{5}}\text{大气压(极大)}$$

称为电子气体的简并压,是泡利不相容原理和电子气体具有高密度的结果. 简并费米气体的压强在恒星演化中起重要作用.

● 费米气体的特征温度

引入费米气体的特征温度 $\boxed{T_F = \dfrac{\mu_0}{k}}$

上面的 μ_0 对应着 $T_F \sim \mathbf{5\times 10^4 K}$, 显然, $\bar{\upsilon}_0, \bar{P}_0, T_F$ 都是相当大的数值

在绝对零度下, Fermi系统的微观状态数: $W_F = 1$

熵: $\boxed{S_0 = k\ln W_F = 0}$

Fermi气处于完全有序的状态

热力学第三定律: 不论是固体还是气体, 都有 $\lim\limits_{T\to 0} C_V = 0$,
故热力学第三定律是量子统计物理学的理论结果.

$S\to 0$ 要求 $W\to 0$, 这意味着在绝对零度下, 粒子只有一种可能的分布, 即处在最低能级态.

- 绝对零度下理想玻色气体与费米气体的区别

  玻色气体：能量、动量、压强、熵均为零

  费米气体：很高的平均能量、动量

  很大的压强

  熵为零

二、有限温度情形的定性分析

通常的温度 $T \ll T_F$

可以想见,这时仅仅是原先处于低于 μ_0 一点点的能级上的粒子有可能被激发到高于 μ_0 一点点的能级上去.

$0 < T \ll T_F$ 时,

$$f(\varepsilon) = \frac{1}{e^{\frac{\varepsilon-\mu}{kT}}+1} \approx \begin{cases} 1, & \varepsilon < \mu - 2kT \\ 0\sim 1, & \mu - 2kT < \varepsilon < \mu + 2kT \\ 0, & \varepsilon > \mu + 2kT \end{cases}$$

注: $T \neq 0$时$\mu \neq \mu_0$, 但是

在$T < T_F$时差别不大.

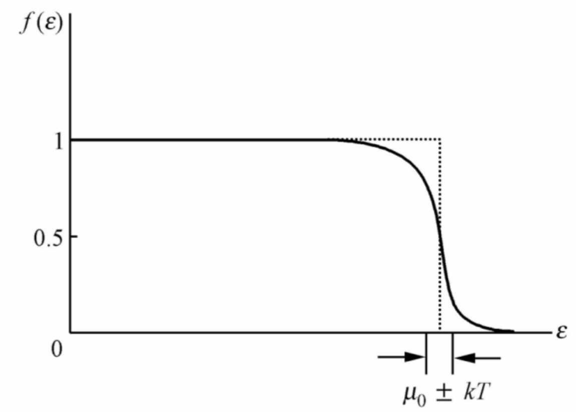


图3-5-2

Fermi气热容量的行为

在温度 T, 粒子分布只在 μ 附近能量宽度约为 $2kT$ 的范围内有变化, 这部分粒子数与总粒子数之比近似为 $2kT/\mu_0$.

由于$kT/\mu_0 \ll 1$, 故被热激发的粒子是总粒子数中很小的一部分. 每个被热激发的粒子具有热运动的平均能量约为kT .

系统能量的增量:

$$\Delta\bar{E} = \bar{E}(T,V) - \bar{E}(0,V) \approx \left(N\frac{2kT}{\mu_0}\right)kT \propto T^2$$

热容量

$$C_V = \left(\frac{\partial\bar{E}}{\partial T}\right)_V \approx 4Nk\left(\frac{kT}{\mu_0}\right) \propto T$$

$C_V \ll (3/2)Nk$ 而且与T成正比(低温费米气体的重要特性之一).

三、有限温度情形的定量计算与分析—简并费米气的宏观性质

- 积分 Q_l

宏观量 $N, \overline{E}$ 以及特性函数 $\Phi(\alpha,\beta,V)$ 的计算中都遇到积分

$$Q_l = \int_0^\infty f(\varepsilon)\varepsilon^l d\varepsilon \qquad (l>0)$$

首先做一次分部积分

$$Q_l = \frac{1}{l+1}\int_0^\infty \varepsilon^{l+1}\left(-\frac{df}{d\varepsilon}\right)d\varepsilon$$

注意 $-(df/d\varepsilon)$ 只在 $\varepsilon \approx \mu$ 的一个狭窄的范围内 $\neq 0$, 所以在 $\varepsilon = \mu$ 附近做展开

$$\varepsilon^{l+1} = \mu^{l+1} + (l+1)\mu^l(\varepsilon-\mu) + \frac{1}{2}l(l+1)\mu^{l-1}(\varepsilon-\mu)^2 + \cdots$$

$$Q_l \approx \frac{1}{l+1}\int_0^\infty [\mu^{l+1} + (l+1)\mu^l(\varepsilon-\mu) + \frac{1}{2}l(l+1)\mu^{l-1}(\varepsilon-\mu)^2]\left(-\frac{df}{d\varepsilon}\right)d\varepsilon$$

$$-\frac{df}{d\varepsilon}=\frac{1}{kT}\frac{\exp\left(\frac{\varepsilon-\mu}{kT}\right)}{\left(\exp\left(\frac{\varepsilon-\mu}{kT}\right)+1\right)^2}$$

做代换 $x=\frac{\varepsilon-\mu}{kT}$

有(准确到 kT/μ 的二次项)

$$Q_l=\frac{1}{l+1}\mu^{l+1}\int_{-\infty}^{+\infty}\left[1+(l+1)\left(\frac{kT}{\mu}\right)x+\frac{1}{2}l(l+1)\left(\frac{kT}{\mu}\right)^2x^2\right]\frac{e^x}{(e^x+1)^2}dx$$

注: 积分下限用 $-\infty$ 代替了 $-\mu/kT \quad (kT\ll\mu)$, 其中

$$\int_{-\infty}^{+\infty}\frac{e^x}{(e^x+1)^2}dx=1;\ \int_{-\infty}^{+\infty}\frac{xe^x}{(e^x+1)^2}dx=0;\ \int_{-\infty}^{+\infty}\frac{x^2e^x}{(e^x+1)^2}dx=\frac{\pi^2}{3}$$

$$Q_l=\frac{1}{l+1}\mu^{l+1}\left[1+\frac{1}{6}l(l+1)\left(\frac{\pi kT}{\mu}\right)^2\right]$$

● 粒子数、化学势

$$N=\int_0^\infty f(\varepsilon)g(\varepsilon)d\varepsilon=CV\int_0^\infty f(\varepsilon)\varepsilon^{1/2}d\varepsilon,\quad \left(l=\frac{1}{2}\right)$$

$$=\frac{2}{3}CV\mu^{3/2}\left[1+\frac{1}{8}\left(\frac{\pi kT}{\mu}\right)^2+\cdots\right]$$

可解出

$$\mu^{3/2}\left[1+\frac{1}{8}\left(\frac{\pi kT}{\mu}\right)^2+\cdots\right]=\frac{3N}{2CV}=\mu_0^{3/2}$$

$$\mu=\mu_0\left[1+\frac{1}{8}\left(\frac{\pi kT}{\mu}\right)^2+\cdots\right]^{-2/3}\approx\mu_0\left[1+\frac{1}{8}\left(\frac{\pi kT}{\mu_0}\right)^2+\cdots\right]^{-2/3}$$

$$\approx\mu_0\left[1-\frac{1}{12}\left(\frac{\pi kT}{\mu_0}\right)^2+\cdots\right]\qquad \left(\frac{\mu_0}{k}\gg T\right)$$

所以 T 增加时 μ 减小, 但 μ 略小于 μ_0, 且两者具有相同的数量级.

【注意】理想费米气体的化学势可正可负, 其数值与气体的温度和粒子数密度有关

$$\begin{cases} T = 0, \quad \mu_0 > 0 \\ T \uparrow, \quad \mu \downarrow \\ T \gg T_F, \text{气体服从半经典分布}, \mu < 0 \\ \qquad (\text{因为} e^{\alpha} = e^{-\frac{\mu}{kT}} \gg 1) \end{cases}$$

- **能量** $$\bar{E}=\int_0^\infty f(\varepsilon)g(\varepsilon)\varepsilon\, d\varepsilon = CV\int_0^\infty f(\varepsilon)\varepsilon^{3/2}d\varepsilon, \qquad \left(l=\frac{3}{2}\right)$$

$$=\frac{2}{5}CV\mu^{5/2}\left[1+\frac{5}{8}\left(\frac{\pi kT}{\mu}\right)^2+\cdots\right]$$

再代入上面的 μ 值, 得

$$\bar{E}=\frac{2}{5}CV\mu_0^{5/2}\left[1-\frac{1}{12}\left(\frac{\pi kT}{\mu_0}\right)^2+\cdots\right]^{5/2}\left[1+\frac{5}{8}\left(\frac{\pi kT}{\mu_0}\right)^2+\cdots\right]$$

$$\approx\frac{2}{5}CV\mu_0^{5/2}\left[1-\frac{5}{24}\left(\frac{\pi kT}{\mu_0}\right)^2+\cdots\right]\left[1+\frac{5}{8}\left(\frac{\pi kT}{\mu_0}\right)^2+\cdots\right]$$

$$\approx\bar{E}_0\left[1+\frac{5}{12}\left(\frac{\pi kT}{\mu_0}\right)^2+\cdots\right]$$

$$\boxed{\bar{E}_0=\frac{2}{5}CV\mu_0^{5/2}=\frac{3}{5}N\mu_0}$$

● 热容量

$$C_V=\left(\frac{\partial \bar{E}}{\partial T}\right)_V=Nk\frac{\pi^2}{2}\left(\frac{kT}{\mu_0}\right)$$

与前面的估算比较，这里系数是 $\pi^2/2\approx 4.9$ **(前面是4).**

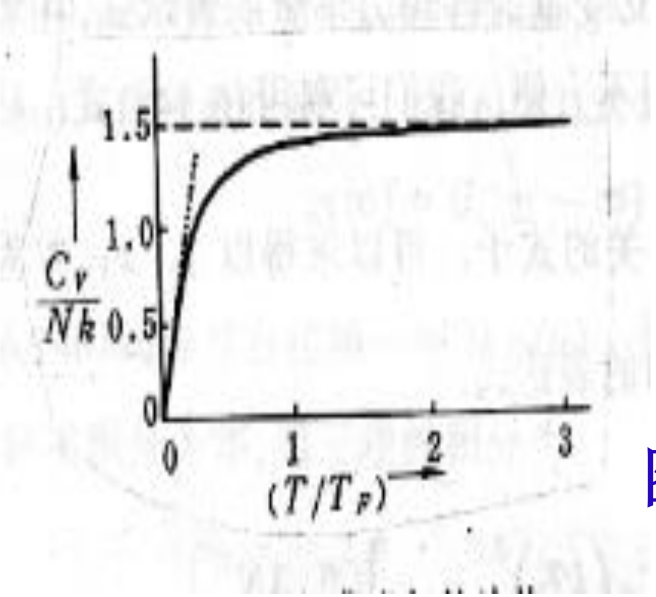


图3-5-3 理想费米气的热容量
点线表示低温下的线性趋势

说明：在常温范围电子的热容量远小于离子振动的热容量.但在低温范围，离子振动的热容量 $C_V\propto T^3$ 而电子热容量 $C_V\propto T$ 减少比较缓慢，所以在足够低的温度下电子热容量不能忽略.

- 特性函数

$$\Phi(\alpha,\beta,V)=\int_0^\infty g(\varepsilon)\ln(1+e^{-\alpha-\beta\varepsilon})d\varepsilon$$

$$=CV\int_0^\infty \ln(1+e^{-\alpha-\beta\varepsilon})\varepsilon^{1/2}d\varepsilon$$

$$=\frac{2}{3}CV\beta\int_0^\infty f(\varepsilon)\varepsilon^{3/2}d\varepsilon \qquad \left(l=\frac{3}{2}\right)$$

$$=\frac{2}{3}CV\beta\frac{2}{5}\mu^{5/2}\left[1+\frac{5}{8}\left(\frac{\pi kT}{\mu}\right)^2\right] \qquad \left(\mu=-\frac{\alpha}{\beta}\right)$$

$$=\frac{4}{15}CV\beta\left(-\frac{\alpha}{\beta}\right)^{5/2}\left[1+\frac{5\pi^2}{8}\left(-\frac{1}{\alpha}\right)^2+\cdots\right]$$

- 压强

$$\bar{P}=\frac{1}{\beta}\frac{\partial\Phi}{\partial V}=\bar{P}_0\left[1+\frac{5}{12}\left(\frac{\pi kT}{\mu_0}\right)^2+\cdots\right]$$

- 熵

$$
\begin{aligned}
S &= k\left(\Phi - \alpha\frac{\partial\Phi}{\partial\alpha} - \beta\frac{\partial\Phi}{\partial\beta}\right) \\
&= k\left(\Phi - \frac{\mu}{kT}N + \frac{1}{kT}\bar{E}\right) \\
&= \frac{\pi^2}{3}CV\mu_0^{1/2}k^2T \\
&= Nk\left(\frac{\pi^2 kT}{2\mu_0}\right) \\
&\propto T
\end{aligned}
$$

四、应用

应用一：金属中的自由电子气

应用二：Pauli顺磁性

应用三：热电子发射

应用四：白矮星(White Dwarf)